1.3.3 Writing Your Own Functions

Show Lecture Video

First, run this code:

```
def cubed(n):
    return n ** 3

print(cubed(2)) # prints 8
print(cubed(3)) # prints 27
```

Run

This code defines a new function named `cubed`. The function takes one parameter, `n`, and returns n^3. The code then calls the function twice.

Important points:

- The first line of a function definition (with the `def`) is the **function header**. It has 4 parts, in order:
  1. The `def` keyword
  2. The **function name**. This is `cubed` in the example.
  3. The **function parameters**, in parentheses. This is `(n)` in the example.
  4. A colon (`:`)
- The rest of the function definition is the **function body**. Note:
  - The body is always indented inside the function definition.
  - The body should contain a `return` statement, which immediately ends the function and returns the result of the call to that function.
  - If the function completes without a `return` statement, then Python automatically returns the value `None`.
- Once a function is defined, you can call it just like any other function.
  - When you call the function, you must provide **arguments**, one for each parameter, in order.

Question: True or False: every function call returns some value?

Show answer

```
def sumOfSquares(x, y):
```

```
def sumOfSquares(x, y):
    return x**2 + y**2

print(sumOfSquares(3, 4)) # prints 25, which is 3
```

Run

This defines a function named `sumOfSquares`. It takes two parameters, `x` and `y`. When we call this function, we must provide two arguments, the values for `x` and `y` in that order.

Functions that Take No Parameters

Here is another example:

```
def bigNumber():
    return 78238477823487248727834

print(bigNumber()) # prints 782384778234872487278
```

Run

Here we define a function that takes no parameters. Note that you still have to include the parentheses, both in the function definition and in the function call.

Functions that Return Multiple Values

Finally, here is one more example:

```
def sumAndProduct(x, y):
    return x+y, x*y

m, n = sumAndProduct(2, 3)
print(m, n)
```

Run

Here we define a function that returns multiple values. Run the code and see how `m` is `2+3`, and `n` is `2*3`.

Checkpoint 1

What are the parameters and arguments in the following code?

```
def f(x, y):
  return x + y

print(f(2, 3))
```

- ○ **`2` and `3` are parameters, and `x` and `y` are arguments.**
- ○ **`2` and `3` are both the parameters and arguments.**
- ○ **`x` and `y` are parameters, and `2` and `3` are arguments.**
- ○ **`x` and `y` are both the parameters and arguments.**

Submit

Checkpoint 2

True or False: The body of a function can only contain one line.

- ○ **True**
- ○ **False**

Submit

Continue